Zhixin Tu

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Education

Southern University of Science and Technology Shenzhen, China

Ph.D. in Intelligent Robotics GPA: 3.61 / 4.0 (Ranking 1 / 14) Aug. 2021 – Jun. 2026

Honors: SUSTech Motto Scholarship Supervisor: Prof. Chenglong Fu

Chongqing University Chongqing, China

B.S. in Mechanical engineering GPA: 3.68 / 4.0 (Top 3%) Sep. 2017 – Jun. 2021

Honors: National Scholarship

Publications

[1] **Z. Tu**, Y. Jiang, H. Yan, Y. Leng, and C. Fu, " Design, Modeling, Control, and Evaluation of a Wearable Centaur Robot for Load-carriage Walking Assistance," in ***International Journal of Robotics Research***, 2025. (Accepted)

[2] **Z. Tu**, H. Liu, Y. Jiang, T. Ye, Y. Qian, Y. Leng, J. Dai, and C. Fu, " Softening Nonlinear Stiffness Elastic Mechanism with Continuous Adjustability for Human–robot Interaction Force Control," in ***Mechanism and Machine Theory***, vol. 200, pp. 105704, Sep. 2024.

[3] **Z. Tu**, Y. Fang, Y. Leng and C. Fu, "Task-Based Human-Robot Collaboration Control of Supernumerary Robotic Limbs for Overhead Tasks," in ***IEEE Robotics and Automation Letters***, vol. 8, no. 8, pp. 4505-4512, Aug. 2023. **(Oral presented on ICRA 2024)**

[4] **Z. Tu**, W. Zhuang, Y. Leng and C. Fu, "Accelerated Informed RRT*: Fast and Asymptotically Path Planning Method Combined with RRT*-connect and APF," in 2024 16th ***International Conference on Intelligent Robotics and Applications***, pp. 279-292.

[5] H. Yan, J. Li, H. Liu, **Z. Tu**, P. Yang, Y. Leng and C. Fu, "Locomotion Control on Human-centaur System with Spherical Joint Interaction," in ***IEEE Robotics and Automation Letters***, vol. 9, no. 5, pp. 4838-4845, May 2024.

Research Experience

- **Legged locomotion control**: Reinforcement learning, model-based optimization
- **Human-robot interaction**: Compliant mechanism, collaborative control
- **Human motion analysis**: Motion intension estimation, walking biomechanics

Project I: Human–Centaur locomotion control via RL and human motion generation

Vision-dependent motion planning often leads to behavioral conflicts between human and Centaur robot. This project aims to develop a robust and adaptive learning locomotion control strategy that ensures human–robot coherence and collaboration during walking on various terrains.

- Developed a human motion priors generation model that captures locomotion patterns across diverse terrains, serving as the foundation for simulating closed-loop human–robot–environment interactions and policy learning.
- Utilized privileged information learning to model the temporal correlations between terrain features and human motion interactions, enabling multi-terrain policy training without explicit perception.

- Proposed a Regularized Online Adaptation framework that mines latent perceptual and decision information from the robot's proprioceptive history, achieving zero-shot transfer to the physical Centaur system. Subsequent studies focus on biomechanical assessments of metabolic efficiency and muscle fatigue across multiple subjects.

Project II: Human–Centaur collaborative loco-interaction control based on compliant decoupling

The Centaur robot is required to maintain balance and generate desired interaction forces under human gait disturbances, achieving coordinated, multidirectional, and variable-speed human-Centaur locomotion control.

- Formulated a compliant interaction dynamics model to decouple the human–Centaur system and derive reduced-order single- and multi-body representations for control strategy.
- Developed a compliant decoupling–based loco-interaction MPC and WBC architecture integrating adaptive weighting, soft contact, and foot configuration planning for stable and coordinated human–robot walking.
- Achieved **52.22%** load sharing and a **35.16%** reduction in metabolic cost during load-carriage walking, representing the best assistance performance reported to date.

Project III: Design of a Centaur system with softening nonlinear elastic coupling

Existing load-carriage assistive robots often exhibit poor human–robot coordination and inefficient locomotion assistance. This project introduces a novel human–robot coupling concept to enable efficient and coordinated load-carriage walking assistance.

- Proposed a human–quadruped walking concept that extends human bipedal morphology into a quadruped configuration, providing combined vertical load sharing and horizontal propulsive assistance.
- Designed the Centaur robot structure to realize front–rear human–robot coupling and optimized load distribution, enhancing overall gait stability and coordination.
- Developed a reconfigurable softening nonlinear elastic mechanism with interaction force control error below 5 N, achieving the first realization of forward propulsive assistance on a mobile wearable robotic system.

Project IV: Human-robot collaborative control of supernumerary robotic limbs

The heterogeneous human–SRL system lacks effective channels for conveying task information and human intent, making seamless cooperation in complex tasks challenging.

- Developed a task model to represent the collaboration process of overhead tasks, extracting human motion intention and task parameters to drive task-state transitions.
- Proposed an adjustable admittance controller that adapts stiffness and damping characteristics to different task states, ensuring interaction safety and reliability.
- Achieved task-adaptive and seamless human–SRL collaborative control, effectively reducing human cognitive load, with task parameter estimation error below 2 cm.

WORKING EXPERIENCE

Reinforcement Learning and Control Research Intern Sep. 2025 – Present

Mondo Robotics, Shenzhen, China Mentor: Shuo Yang

- Responsible for developing reinforcement learning–based locomotion control algorithms for wheel–legged robots, including robust high-speed locomotion, jumping, and fall-recovery strategies across multiple terrains. Conducted sim-to-sim and sim-to-real deployment validation to ensure consistency real-world performance.

Other Information

Technical skills:

- Robotic System Development: Familiar with the full R&D pipeline from mechanical design and system integration to control implementation. Led the mechanical design, assembly, and debugging of robotic systems, as well as the development of both low-level and high-level control algorithms.
- Robot Control: Proficient in robot kinematics and dynamics modeling, and experienced in admittance/impedance control, model predictive control (MPC), and reinforcement learning–based locomotion control for legged robots.
- Programming & Development Tools: Skilled in Python, C/C++, and MATLAB; proficient with PyTorch, ROS, and Linux environments; familiar with CAN bus and serial communication protocols.
- Simulation Platforms & Software: Experienced with major robotics and simulation tools, including Isaac Gym/Lab, MuJoCo, Webots, SolidWorks, and Adams
- Team and Project Management: Experienced in coordinating collaborative research projects; Led a research group of six M.S./Ph.D. students and supervised undergraduate thesis projects over several years.

Research funding participation:

- Served as a key contributor in several national-level research funding applications, including projects supported by the National Natural Science Foundation of China (NSFC) and the Ministry of Science and Technology (MOST) Key R&D Program.
- Assisted in grant proposal writing, technical documentation, and presentation material development, gaining solid experience in research funding preparation and submission processes.

Academic service:

- Serve as a reviewer for reputable journals (*IJRR, T-RO, T-ASE, RAL, etc.*) and flagship robotics research conferences (*ICRA, IROS, etc.*).

Languages:

- English (working proficiency), Chinese (native).